

SADRODIN BARIKBIN

Tehran, Iran

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EDUCATION

Sharif University of Technology, Tehran, Iran

September 2022

M.Sc. in Artificial Intelligence & Robotics

Thesis: Improving Pre-trained Language Model for Sentiment Analysis

University of Tehran, Tehran, Iran

September 2019

B.Sc. in Computer Engineering: Software Engineering

EXPERIENCE

BIMLOGIQ

April 2025 - Present

AI Software Developer

Remote

- Helped build **Argus**, the company's new Revit AI agent (Pydantic AI on AWS Bedrock AgentCore), working full-stack across its Next.js frontend, Python backend, and C# Revit plugin.
- Added real-time, cancellable streaming to Argus's Revit code execution (SSE progress, mid-run cancellation, timeouts) across the C# plugin, Python backend, and React frontend.
- Built Argus's Microsoft 365 (Graph, OAuth On-Behalf-Of) and Autodesk Platform Services (11 tools, 3-legged OAuth) integrations, plus Excel and PDF file-processing tools.
- Shipped a command-in-chat feature end-to-end (Python backend, C# plugin, AngularJS UI) for **Copilot**, Argus's predecessor.
- Researched Copilot's Revit entity-retrieval indexing (Lucene.NET vs. vector search), and built custom C# Roslyn analyzers that deterministically catch errors in agent-generated code.

Open-Source Contributions

February 2022 - Present

- PyTorch-Ignite Core Developer (emeritus)**: 58 merged PRs (2022–2024) adding features (a MetricGroup API, a COCO mAP metric, serializable metrics) and distributed-training fixes.
- 60+ further merged PRs across the ML/Python ecosystem, mostly small fixes and documentation: pytest, HuggingFace Transformers and LightEval, LangChain, Pydantic-AI, scikit-learn, NetworkX, and CPython.

BaSalam

October 2024 - March 2025

AI Developer

Qom, Iran

- Fine-tuned multiple LLMs (LoRA, base models up to 4B) for user query expansion; evaluated variants with offline relevance metrics.
- Built an internal evaluation app for human-in-the-loop comparison; automated deployment and inference with vLLM for rapid iteration.

CRCIS

October 2023 - May 2025

AI Software Developer

Qom, Iran

- Improved and evaluated a Tesseract-OCR pipeline (pre-processing, post-processing, and error analysis for Persian text).
- Built an LLM leaderboard to evaluate Quranic knowledge in Persian; designed prompt sets and evaluation protocols.
- Fine-tuned an embedding model for Hadith retrieval and validated ranking quality on curated queries.

Sahmeto

June 2022 - November 2022

Machine Learning Intern

Tehran, Iran

- Fine-tuned a language model on crypto market data to automate labeling and reduce manual review effort.

Data Analytics Lab at University of Tehran
C++ Developer

June 2019 - February 2020
Tehran, Iran

- Modified cryptocurrency clients to stream blockchain data into MongoDB and Neo4j in analysis-friendly formats.
- Built indexes and precomputed summaries to accelerate analytics workflows on large blockchain datasets.

TECHNICAL SKILLS

Languages: Python, C#, C++, Java, JavaScript/TypeScript
ML/NLP: PyTorch, Transformers, LoRA, vLLM, Pydantic AI, IR (Lucene.NET), OCR (Tesseract)
Data/Backend: FastAPI, PostgreSQL, MongoDB, Neo4j, Docker, AWS (Bedrock/Lambda), Sentry, Linux
Frontend: React/Next.js, AngularJS, VueJS, ElectronJS
AI Dev Tools: Claude Code and other AI coding agents

PUBLICATIONS

ICML 2023 Topological Deep Learning Challenge: Design and Results
Papillon, Mathilde et al.
Proceedings of 2nd Annual Workshop on Topology, Algebra, and Geometry in Machine Learning (TAG-ML). 2023

Making Use of Pre-Trained Model’s Attention Values in Structured Sentiment Analysis
Barikbin, Sadrocin
Proceedings of the 16th International Workshop on Semantic Evaluation (SemEval-2022). 2022

CERTIFICATIONS & AWARDS

Most efficient method award in CLAS2024, Backdoor trigger recovery for LLM Agents *October 2024*
Neurips 2024 (TulipTrackers team: Ali Satvaty, Anna Visman, Sadrocin Barikbin)

2nd place winner in ICML 2023 Topological Deep Learning Challenge *October 2023*
In Hypergraph domain

Deep Reinforcement Learning Course *May 2023*
Issued by HuggingFace

Ranked 9th in National Entrance Exam for Graduate Studies in Computer Science Major *October 2019*
Issued by National Organization of Educational Testing, Tehran, Iran

Arduino Workshop *August 2017*
Issued by Shahid Beheshti University IEEE Student Branch, Tehran, Iran

Ranked among top 0.5% of competitors in National University Entrance Exam *September 2015*
Issued by National Organization of Educational Testing, Tehran, Iran

LANGUAGES

English *Fluent*
Persian (Farsi) *Native*
Arabic *Intermediate*
French *Basic*